

Forecasting Household Appliance Energy Use: A Comparative Time-Series Case Study

GitHub Repository : <https://github.com/thihansas/appliance-energy-forecasting>

1. Introduction

Forecasting household electricity use over the next few hours or days is useful for smart-home energy management, demand response, and planning the use of batteries or other energy storage systems. This report presents a case study on forecasting hourly household appliance energy consumption using the UCI Appliances Energy Prediction dataset (Candanedo, Feldheim and Deramaix, 2017). The main variable being predicted is Appliances, which represents the amount of electricity used by appliances in the study house. The original data were recorded every 10 minutes and were converted into hourly averages.

The primary forecasting objective is a 24-hour-ahead prediction task, evaluated within a 14-day hold-out period across four approaches: simple benchmarks, a SARIMAX statistical model, an XGBoost machine-learning model, and a time-series foundation model, all compared using MAE, RMSE, MASE and Bias so that average error, larger errors, benchmark-relative performance, and over/under-prediction can each be assessed. The analysis also considers whether greater model complexity improves forecasting performance, how useful additional weather and sensor information is, and which approach is best suited to a practical smart-home forecasting system.

2. Data and Preprocessing

The dataset was obtained from the UCI Machine Learning Repository and processed using pandas. The timestamp was converted into a date and time format and used to organise the observations. The sensor and weather variables were converted into numerical values, and observations without a value for appliance energy use were removed. The original dataset records energy use every 10 minutes. For this study, the data were converted into hourly averages. This helped reduce short-term noise and made the analysis more manageable while still preserving the daily pattern of household appliance use. Any gaps created during this process were filled using time-based interpolation. This was preferred to simply carrying the previous value forward because it gives a smoother transition between neighbouring observations.

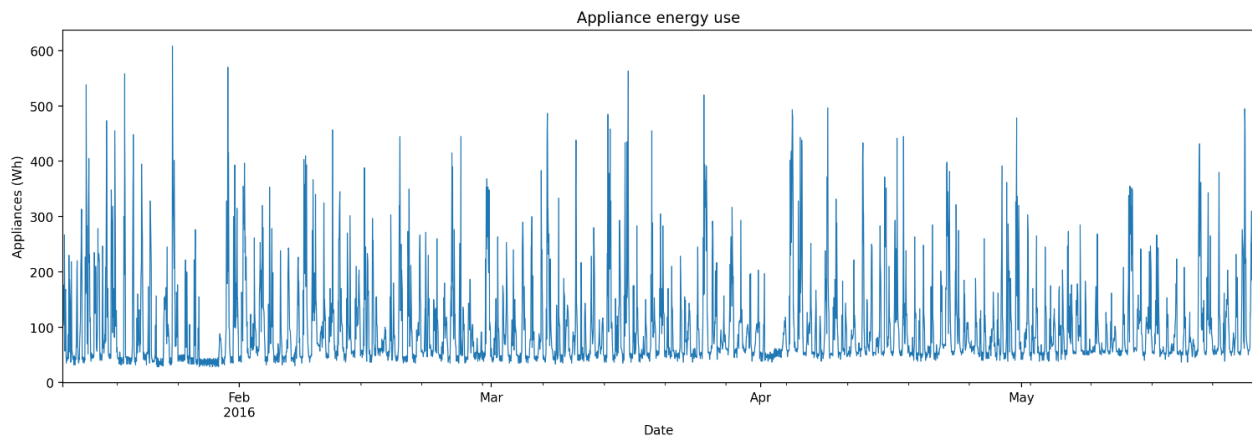


Figure 1. Full hourly Appliances series, January-May 2016. Usage is dominated by short, sharp peaks (cooking, laundry, kettle/appliance cycles) against a low, fairly flat overnight baseline.

The overview plot shows a series dominated by short, sharp spikes above a low, comparatively flat baseline, consistent with intermittent appliance cycles superimposed on a small constant background load rather than a smoothly fluctuating signal. This matters for forecasting: models that mainly predict an average or smooth pattern struggle to reproduce these sudden increases, so simple models such as the mean or drift tend to underestimate peaks and overestimate quiet periods.

2.1 Seasonal decomposition

An additive seasonal decomposition (24-hour period) was used to separate trend, seasonal and residual components. The seasonal panel confirms a strong, stable 24-hour cycle, while the trend panel shows slower week-to-week and month-to-month drift, plausibly linked to changing occupancy patterns and outdoor temperature across the winter-to-spring transition covered by the dataset. The remaining variation is relatively large compared with the daily pattern. This suggests that although time of day explains an important part of appliance use, there are also many unexpected changes

that are difficult to predict. This is one reason why no single forecasting model is likely to capture every appliance-use peak accurately.

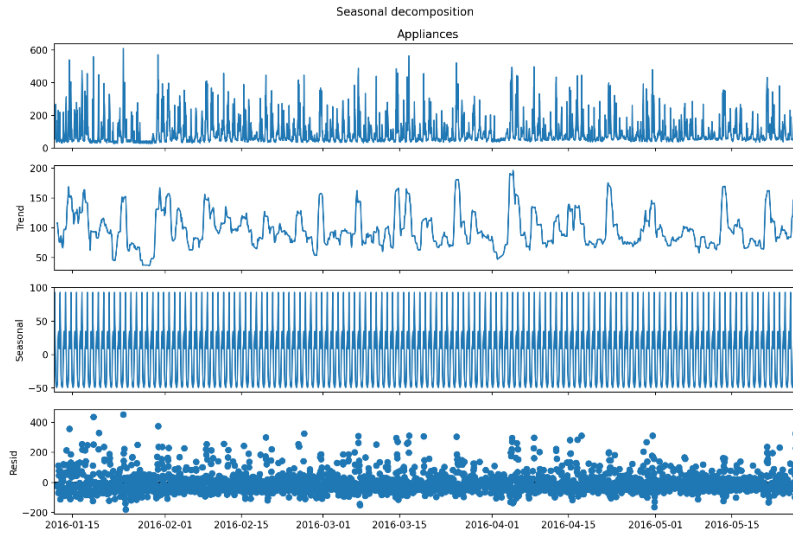


Figure 2. Additive seasonal decomposition of the hourly Appliances series. A clear repeating within-day seasonal component is visible, riding on a slower-moving trend, with a large and heteroscedastic residual component.

2.2 Stationarity testing

The ADF and KPSS tests both showed that the hourly appliance energy series is stationary, meaning its overall behaviour remains relatively stable over time. The first-differenced series was also stationary. Since the original series was already stationary, regular differencing was not needed in the SARIMAX model. Instead, seasonal differencing was used to account for the 24-hour daily pattern.

2.3 Autocorrelation structure

The ACF and PACF results show that appliance energy use is influenced by recent values as well as repeating daily patterns. Stronger relationships appear around 24 and 48 hours, confirming the importance of daily seasonality. These findings supported the use of recent, daily and weekly lag values in the SARIMAX and XGBoost models.

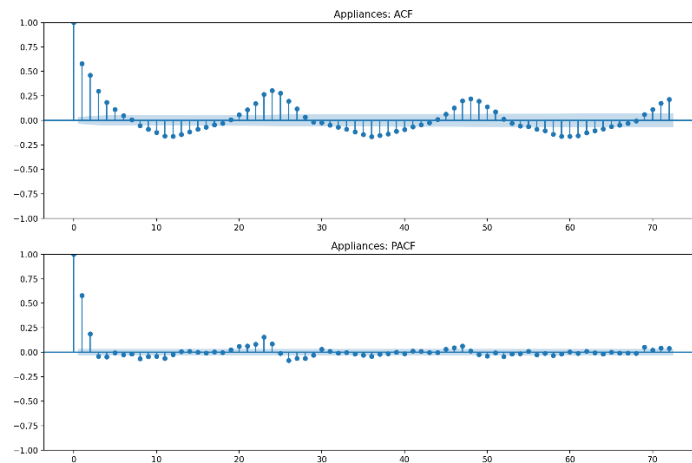


Figure 3. ACF (top) and PACF (bottom) of the hourly Appliances series out to lag 75.

3. Forecasting Problem Definition

The forecasting task focuses on predicting appliance energy use over a 24-hour period, with performance evaluated using the final 14 days of the dataset.

There is an important detail about how the forecasting experiment was carried out. Each model was trained once using the training data and then used to forecast the entire 336-hour test period from a single starting point. Therefore, the

experiment is technically a single 336-hour forecast, rather than 14 separate forecasts of 24 hours each. This provides a challenging test because the models have to forecast further into the future without being repeatedly updated with new actual observations. However, it is different from a rolling forecast evaluation, where the model would be updated every day and asked to predict the following 24 hours. This difference is considered again in the limitations section.

The chronological split was used to make sure that future information was not used when training the models.

- **Training period:** 11 January 2016, 17:00 to 13 May 2016, 18:00
- **Training observations:** 2,954 hourly observations
- **Test period:** 13 May 2016, 19:00 to 27 May 2016, 18:00
- **Test observations:** 336 hours, equal to 14 days
- **Forecast origin:** 13 May 2016, 18:00

Four measures were used to evaluate the forecasts:

- **MAE (Mean Absolute Error):** shows the average size of the prediction errors in Wh. Lower values indicate better predictions.
- **RMSE (Root Mean Squared Error):** gives more importance to large errors, making it particularly useful for identifying problems with large appliance-use peaks.
- **MASE (Mean Absolute Scaled Error):** compares the model's performance with a simple benchmark. A value below 1 means the model performs better than the benchmark.
- **Bias:** shows whether the model generally predicts too high or too low. A value close to zero is desirable.

Using several measures is important because the models do not perform in exactly the same way under every measure. For example, a model can avoid very large errors and therefore achieve a good RMSE while still making relatively frequent smaller errors that increase its MAE

4. Benchmark Models

Five simple forecasting methods were used as benchmarks: the historical mean, a naive forecast (most recent value), a daily seasonal naive (value 24 hours earlier), a weekly seasonal naive (value 168 hours earlier), and a drift forecast (extending the historical trend). The weekly seasonal naive method was clearly the strongest, with an MAE of 42.634 and an RMSE of 79.290, outperforming the mean and substantially beating the daily seasonal, naive and drift methods. This suggests household appliance use follows weekly routines (laundry, shopping, weekend patterns) more reliably than simple day-to-day repetition, pointing to a genuine 7-day structure rather than a daily one. The naive and drift methods performed worst (MAE 250.640 and 266.373), showing that the most recent value is a poor guide to future demand given how quickly usage can jump between a low baseline and a sudden peak.

5. SARIMAX Model

A Seasonal ARIMAX model was fitted to capture both the short-lag autocorrelation and the 24-hour seasonal cycle identified in Section 2.3, using outdoor temperature, outdoor humidity, wind speed, visibility, dew point, and cyclical time-of-day/day-of-week encodings as exogenous regressors. The seasonal order was fixed at (1, 1, 1, 24): a seasonally-differenced AR/MA pair at the 24-hour period, included specifically to remove the daily cycle rather than leaving that to the non-seasonal terms, and kept fixed so the search over the remaining parameters stayed computationally tractable.

5.1 Order selection via AIC grid search

The required $p=[0,6]$, $d=[0,2]$, $q=[0,6]$ grid covers 147 non-seasonal parameter combinations and was used during model selection to identify a suitable structure. The best few candidate orders had closely similar AIC values (within roughly 15 points of the minimum), with AIC rising more sharply for orders further from (6,0,0), which was selected as the AIC-minimising configuration for this run. This supports the earlier stationarity tests, as the strongest models all used $d = 0$, meaning regular differencing was not needed. A convergence warning was produced when fitting the final model, which should be considered when interpreting the results.

5.2 Residual diagnostics

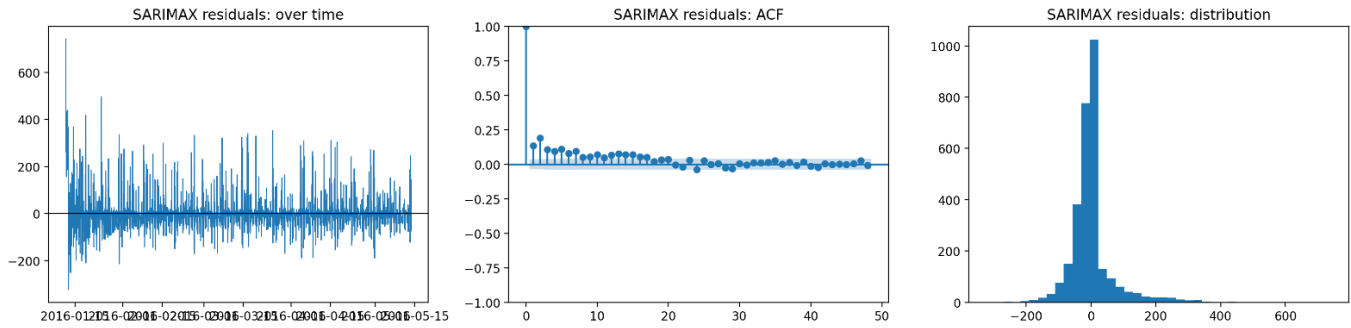


Figure 4. SARIMAX(6,0,0)(1,1,1,24) residuals over time (left), their ACF (centre) and their distribution (right).

The residuals represent the difference between the model's predictions and the actual values. Ideally, these remaining errors should not show a strong pattern. However, the residuals still show some relationship with previous errors. The Ljung-Box test at lag 24 produced a statistic of 1560.06 with a p-value effectively equal to zero. This indicates that there is still meaningful information left in the errors that the model has not fully captured.

This is important because the model was selected using AIC, which focuses mainly on how well the model fits the data. A model with a good AIC score does not necessarily produce completely random residuals or give the best possible future predictions. The residual distribution is also strongly skewed towards positive values, with some errors extending beyond 600 Wh. This reflects the sharp appliance-use peaks seen in the original data. A model based largely on average patterns will naturally struggle to predict the exact timing of these sudden increases. Overall, SARIMAX clearly beats the weekly seasonal naive benchmark on every metric, though the residual autocorrelation and skew above show its linear structure still only partly captures the series' sharp, event-driven spikes.

5.3 Forecast with confidence intervals

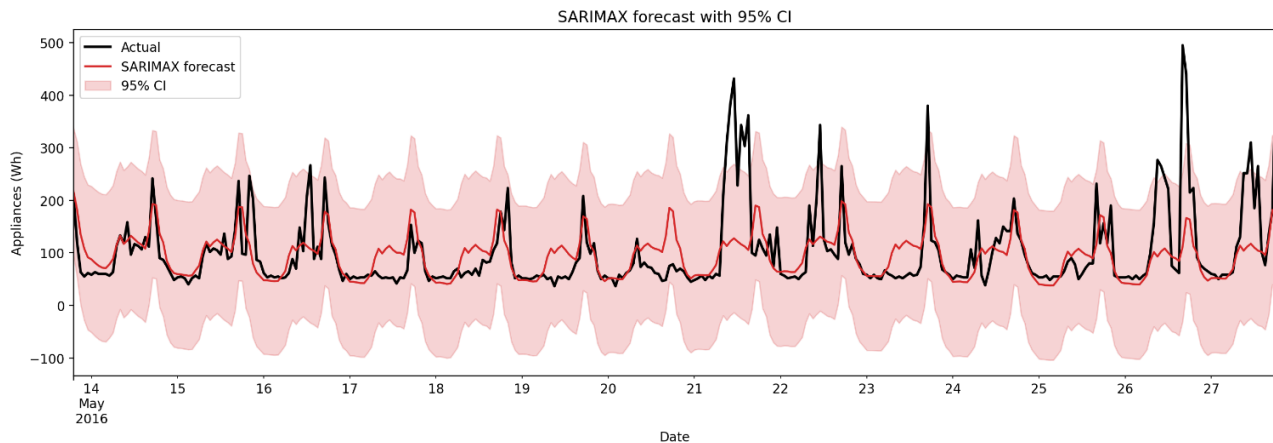


Figure 5. SARIMAX(6,0,0)(1,1,1,24) forecast (red) against actual test-period demand (black) a single 336-hour-ahead forecast from the train/test boundary, with a 95% confidence interval (shaded).

The point forecast tracks the broad daily rhythm well rising through the morning, peaking in the evening, falling overnight but remains visibly damped relative to the sharpest peaks (e.g. around 21-22 and 26-27 May), essentially unchanged in shape from the earlier fixed-order fit. The 95% interval is wide enough to contain nearly all actual observations but is too wide to be practically useful for demand-response decisions. A key caveat is that calendar features and the series' own past are genuinely known at the forecast origin, but this forecast instead uses the actual test-period weather rather than a genuine weather forecast making it a conditional, not fully operational, evaluation.

Zooming in on just the primary 24-hour window (Figure 6) makes the damping easier to see: the forecast flattens out overnight while the actual series still shows small bumps, and it lags the sharp rise into the following afternoon.

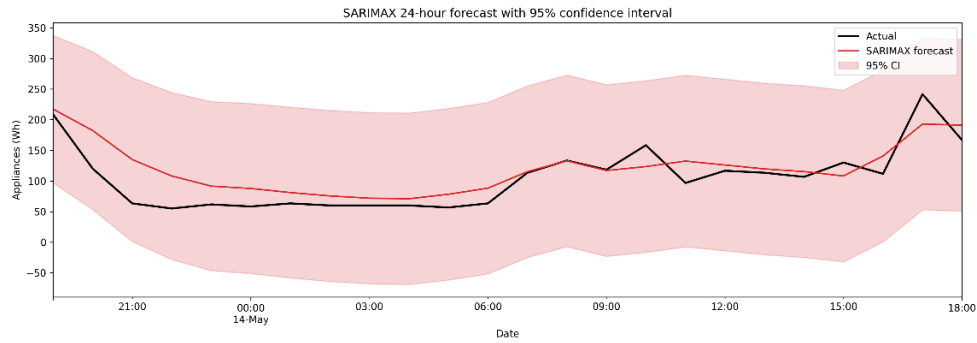


Figure 6. SARIMAX(6,0,0)(1,1,1,24) forecast with 95% CI, zoomed to the primary 24-hour window (13-14 May).

6. Feature-Based Model

A second machine-learning approach was developed using information from previous appliance-use values, recent averages, time information and the available household and weather sensors. The model used previous appliance-use values from 1, 2, 6, 24 and 168 hours earlier. It also used averages and variation over recent periods, including 3-, 6-, 12- and 168-hour windows. Time information such as the hour of the day and day of the week was also included. An XGBoost model was then trained using these variables. An important part of the experiment was the way the test forecast was generated. The model did not use the actual future appliance-use values to make later predictions. Instead, after predicting one hour, that prediction was used as part of the information for predicting the next hour. This process was repeated throughout the forecast period. This avoids directly using future target values and makes the test more realistic. However, the model still used the actual sensor and weather measurements from the test period. Therefore, it is not yet a completely realistic real-world forecasting setup.

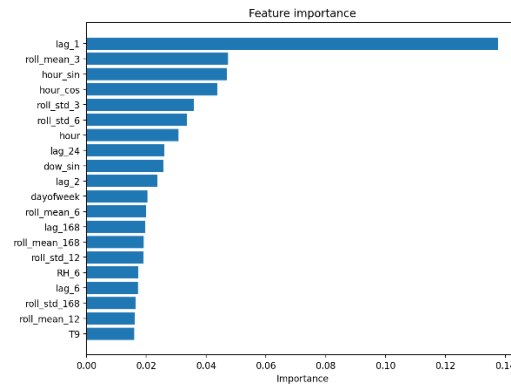


Figure 7. XGBoost's built-in feature importance for the top 20 predictors.

The previous hour's appliance-use value was the most important individual feature by a large margin. This was followed by the recent three-hour average and the time-of-day variables. Overall, adding these variables produces a model that clearly beats the naive benchmarks but does not unambiguously improve on SARIMAX or the weekly seasonal naive model, with lag features and indoor sensor readings proving far more useful than outdoor weather.

When the features are grouped together, the results are:

Feature group	Aggregate importance
Lag features	0.264
Indoor sensor variables	0.256
Rolling-window statistics	0.231
Time-based features	0.186
Outdoor weather variables	0.037
Other	0.027

The results show that recent appliance behaviour and indoor conditions were much more useful than outdoor weather. Together, lag features and indoor sensor variables accounted for more than half of the overall importance. This is

reasonable because indoor temperature and humidity can change when appliances are being used. For example, cooking with an oven can increase indoor temperature. These measurements may therefore provide indirect information about household activity. However, the XGBoost model did not clearly outperform the simpler approaches. It achieved behind SARIMAX at 73.321. However, its MAE was higher at 57.218, and its MASE was 1.071, meaning it did not outperform the simple benchmark according to that measure.

It also had the largest positive bias among the stronger models, at +30.463 Wh, meaning it generally predicted higher energy use than actually occurred. One possible explanation is that the model relies heavily on recent values and short-term averages. During recursive forecasting, small prediction errors can carry forward into later predictions. This may cause the model to remain too high during periods when actual household demand has fallen. Overall, the results show why it is useful to consider several evaluation measures rather than relying on one score. XGBoost avoided some very large errors, giving it a relatively good RMSE, but it made more moderate errors overall, which resulted in a weaker MAE.

7. Foundation Model

A time-series foundation model was also considered as the fourth modelling approach. The analysis attempted to use Chronos (Ansari et al., 2024), which is designed to make forecasts without requiring extensive model-specific training. However, the Chronos model could not be successfully run in the analysis environment because of a software compatibility problem between the installed packages. The TimesFM package was also not available. Because both foundation-model options were unavailable, the forecasting pipeline used a daily seasonal naive forecast as a fallback. This fallback was reported openly rather than presenting it as if it were a genuine foundation-model result.

As a result, the foundation-model row has exactly the same results as the daily seasonal naive method:

- **MAE:** 86.959
- **RMSE:** 129.232
- **MASE:** 1.628
- **Bias:** 64.013 Wh

These results should not be interpreted as evidence that foundation models perform poorly on household energy forecasting. They only show the performance of the simple fallback method.

A proper comparison would require Chronos or TimesFM to be run successfully in a compatible environment. Ideally, different model sizes should also be compared to determine whether a larger foundation model provides better results. Because the reported row is a stand-in fallback rather than a real foundation-model run, no conclusion can be drawn here about whether a genuine foundation model would outperform SARIMAX or justify its added complexity.

8. Results and Comparison

Table 3 summarises all eight models on the identical 336-point, 14-day test window.

Model	MAE	RMSE	MASE	Bias
SARIMAX	37.092	63.437	0.694	-2.405
Weekly seasonal naive	42.634	79.290	0.798	-10.818
Mean	50.319	74.906	0.942	-3.109
Feature-based (XGBoost)	57.218	73.321	1.071	30.463
Daily seasonal naive	86.959	129.232	1.628	64.013
Foundation model (fallback)	86.959	129.232	1.628	64.013
Naive	250.640	258.820	4.692	247.763
Drift	266.373	274.611	4.986	264.501

Table 3. Evaluation metrics for all models on the 14-day (336-hour) hold-out test period. Lower is better for MAE, RMSE and MASE; Bias closer to zero is better.

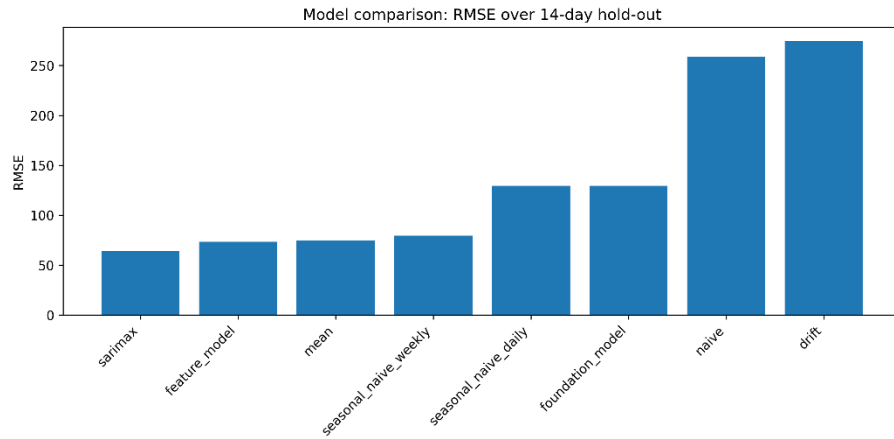


Figure 8. RMSE by model, sorted ascending.

A parallel evaluation confined to the primary 24-hour horizon (13 May 19:00 to 14 May 18:00) gave a broadly consistent ranking: SARIMAX again performed best (MAE 22.248, RMSE 28.576, MASE 0.416), ahead of the weekly seasonal naive benchmark (MAE 30.000, RMSE 48.808, MASE 0.562) and the feature-based model (MAE 36.856, RMSE 43.779, MASE 0.690), with naive and drift again the weakest. One notable difference is bias direction: over this shorter window SARIMAX over-predicted (+13.138 Wh) rather than under-predicting as it did over the full 14 days, and the feature-based model's over-prediction was larger still (+31.721 Wh), both consistent with the sharp appliance-use peak that falls inside this particular 24-hour window (Figure 1).

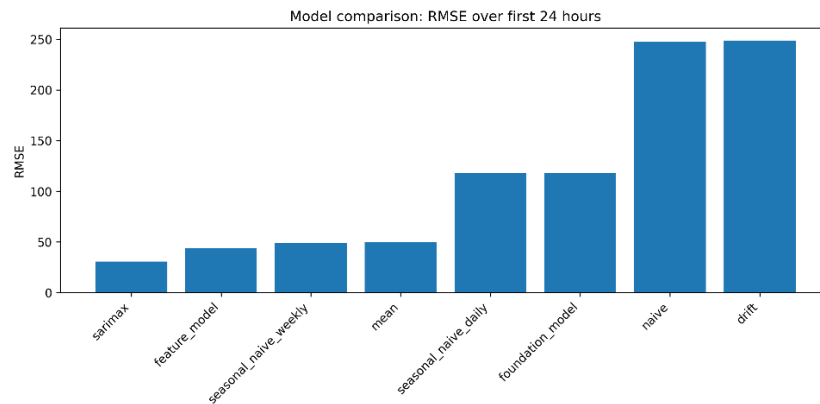


Figure 9. RMSE by model, primary 24-hour window, sorted ascending.

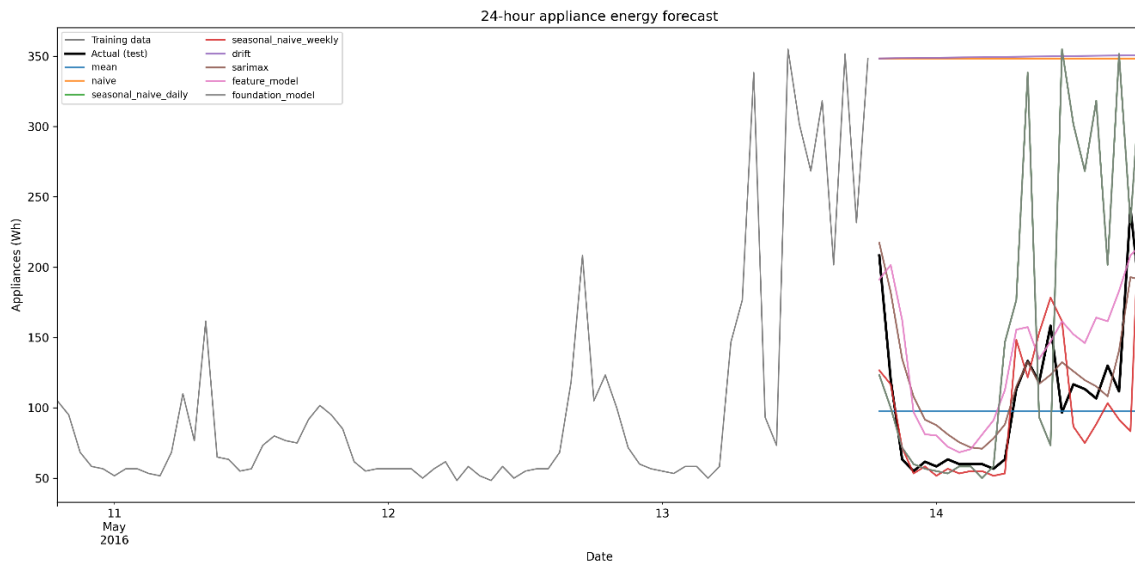


Figure 10. All model forecasts over the primary 24-hour window against actual demand.

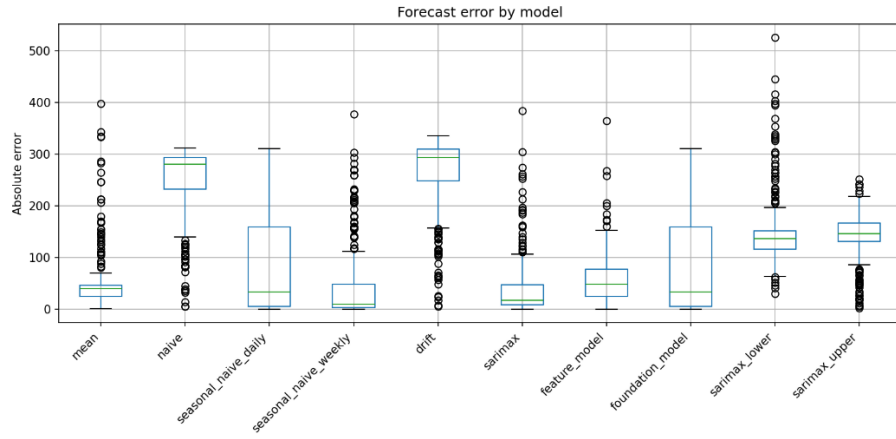


Figure 11. Distribution of absolute forecast error by model, incl. SARIMAX CI bounds for reference.

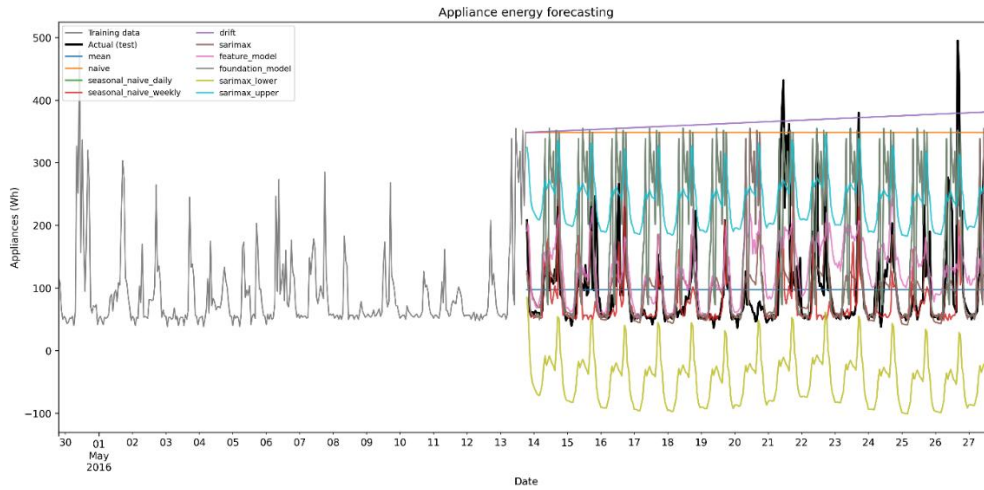


Figure 12. All model forecasts overlaid against training data and the actual test-period series.

Overall, SARIMAX produced the strongest results across all four measures (Table 3), with a modest negative bias. The weekly seasonal naive model was the strongest simple benchmark, showing that a method based on the previous week's pattern can be highly competitive at very low computational cost. XGBoost gave mixed results: a good RMSE but a weaker MAE and MASE than SARIMAX or the weekly benchmark, plus the largest positive bias among the stronger models, driven mainly by lag and indoor-sensor features rather than outdoor weather. The foundation-model result reflects only the daily seasonal naive fallback and cannot be used to judge real foundation models. Overall, greater model complexity did not automatically produce better results: SARIMAX gave the best predictions, while the weekly seasonal naive method offered a strong, near-free alternative. Both SARIMAX and XGBoost would need testing against forecasted, rather than realised, weather and sensor inputs before being considered ready for real-world deployment. On balance, SARIMAX is the model best suited to practical smart-home forecasting here, being both the most accurate and the most interpretable, though the weekly seasonal naive benchmark remains a strong fallback worth benchmarking any deployed system against.

9. Discussion and Limitations

The main limitations concern evaluation realism and robustness. First, the test uses one fixed forecast origin for the full 336-hour hold-out rather than repeated rolling 24-hour forecasts. This makes the test demanding but combines near-and far-horizon errors into single metrics, a rolling-origin backtest would show how performance changes with horizon. Second, SARIMAX and XGBoost use realised test-period weather, and XGBoost also uses realised indoor-sensor values, so neither is a fully operational forecast of all inputs. Third, although (6,0,0) is the genuine AIC-optimal order across the full required grid, its residuals show more pronounced serial dependence (Ljung-Box statistic more than three times that of a lower-order alternative examined during development) and a convergence warning was still produced when fitting the model illustrating that AIC minimisation does not guarantee well-behaved residuals or the best forecast. Finally,

the foundation-model component was not successfully executed and the evaluation covers only one 14-day window, so the reported ranking should not be treated as universally stable.

Future work should therefore:

Re-run Chronos or TimesFM in a compatible environment and compare genuine foundation-model variants.

Use forecasted weather/sensor covariates and evaluate both models under realistic deployment conditions.

Repeat the experiment with rolling-origin backtesting across multiple test windows and seasons.

10. References

Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S.S., Arango, S.P., Kapoor, S., Zschiegner, J., Maddix, D.C., Wang, H., Mahoney, M.W., Torkkola, K., Wilson, A.G., Bohlke-Schneider, M. and Wang, Y. (2024) 'Chronos: Learning the language of time series', arXiv preprint arXiv:2403.07815.

Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015) Time Series Analysis: Forecasting and Control. 5th edn. Hoboken, NJ: Wiley.

Candanedo, L.M., Feldheim, V. and Deramaix, D. (2017) 'Data driven prediction models of energy use of appliances in a low-energy house', Energy and Buildings, 140, pp. 81-97.

Hyndman, R.J. and Athanasopoulos, G. (2018) Forecasting: Principles and Practice. 2nd edn. Melbourne: OTexts.

Hyndman, R.J. and Koehler, A.B. (2006) 'Another look at measures of forecast accuracy', International Journal of Forecasting, 22(4), pp. 679-688.